

Management and Computer Science

Finance and Fintech

Financial Report for Alphabet Inc. and Netflix Inc.

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Introduction

This report provides a financial analysis of Alphabet Inc. and Netflix Inc., examining key financial metrics and market valuations in their market sectors. We compute their Price-to-Earnings ratios compared to similar companies, Discounted Cash Flow valuations, and bond pricing to assess their investment potential and financial health. The report also delves into their capital structures, dividend policies, and beta values to evaluate their risk profiles and market behavior. This analysis aims to provide a financial understanding of the companies using the concepts and models that have been provided in the course.

To see the computations performed and the figures shown in this report check the [Appendix](#).

Netflix, Inc., co-led by Ted Sarandos and Greg Peters, is a firm active in the streaming services sector for its library of films and series, including original content. With a [market cap](#) of approximately \$145 billion, Recent activities, such as the significant sale of shares by Co-CEO Gregory Peters, highlight proactive management and provide key insights for investors regarding leadership confidence and valuation perspectives. ([Yahoo Finance](#))

Alphabet Inc., under the management of CEO Sundar Pichai since December 2019, maintains a focus on technological innovation and digital advertising. With a [market cap](#) of around \$1.98 trillion, Alphabet's strategic advancements into AI and cloud computing have notably expanded its influence across tech sectors, gathering positive investor feedback due to sustained growth and innovation efforts. ([Yahoo Finance](#))

Financial Analysis

Stock Valuation by Comparables

In our analysis, we assessed the intrinsic value of the stock prices for Netflix and Alphabet using valuation by comparables. This approach involves analyzing similar companies within the same industry to make a relative valuation of a company's stock. By comparing financial metrics such as Price-to-Earnings (P/E) ratios of our companies with the industry's average, we derived insights into whether the stocks are undervalued, fairly valued, or overvalued relative to the market.

For Alphabet, we have chosen for comparison: Microsoft, Apple, and Meta. These companies are competitors in the technology sector, engaging in various activities from software development and hardware to social media and digital advertising. In the case of Netflix, we have selected Paramount, Fox, and Disney as comparable companies. These firms represent significant entities in the media and entertainment industry.

Alphabet's P/E ratio (37,01), being lower than the industry average (44,70), suggests that it might be undervalued relative to its peers Apple, Microsoft and Meta. This could suggest a buying opportunity if other factors like growth prospects, market position, and financial health

are favorable. A lower P/E ratio can imply that the market expects slower growth or lower returns from Alphabet compared to the industry average. This needs to be contextualized with Alphabet's specific business strategy, upcoming products, or market changes.

Netflix's P/E ratio (69,00) is way higher than the industry average (19,13) which suggests that investors expect higher earnings growth in the future compared to peers. This could indicate that Netflix's stock is overpriced. However, Netflix is known for its aggressive growth in the streaming sector and is considered to have significantly higher growth prospects than traditional media companies like Fox, Paramount, and Disney, which, along Netflix's business model, could explain such a high P/E value.

Using the last year's EPS for the computation of the P/E ratios, the estimated stock price of Alphabet (\$202,01) is higher than the estimate based on the 5-year average EPS (\$187,83), and for Netflix the estimated price (\$290,75) is also higher with a substantial difference compared to the price estimated with the 5 years average (\$171,18) (See [Appendix](#) for the figures and the computations). Using the 5-year average EPS gives a more stable view of the company's earnings capacity. However, this may not reflect recent changes in the business environment, new revenue streams, or shifts in the company's operational efficiency and may be affected by long-lasting irregular events such as COVID-19. Using the last year's EPS can reflect the company's most current earning power and may be more sensitive to recent market conditions, strategic decisions, or competitive dynamics. It could also be more representative given the unusual economic conditions precipitated by the COVID-19 ([Goldstein, Koijen, & Mueller, 2021](#)) pandemic which lasted for a substantial period of the last five years. Therefore, this method may yield a valuation more compatible with current and forward-looking market state. This explains the expected share price of Alphabet being more undervalued using last year's EPS and the price of Netflix being less overvalued.

Discounted Cash Flow (DCF) Valuation

The Discount Factor Model is used to assess the stock prices of the two companies. We took the free cash flow of the two companies from the last five years' balance sheets, and we assessed the average yearly growth rate which amounted to 25% for Alphabet and 109% for Netflix. Netflix's high growth average is mainly due to the COVID-19 outbreak in which Netflix has very unstable performance with free cash flows having big fluctuations. Then, the free cash flows for the next five years are estimated. Due to the unusual nature of growth rates, expected future cash flows were taken from Bloomberg instead of being assessed with the average growth. The expected values provided in Bloomberg consider past anomalous events and the strategic plans the companies have, which makes a more precise assessment of future free cash flows. Next, the terminal value is computed using the last year's estimated free cash flow, the perpetual growth rate which we assumed to be 3%, and the discount factor which we estimated with the Weighted Average Cost of Capital (WACC).

The values are calculated using the data from the financial statements of the companies. The free cash flows and the terminal value are discounted to the present, then summed. After subtracting the difference between the total debt of the company and the cash/cash equivalence

we get the equity value, which is then divided by the number of shares outstanding to get the estimated stock price of each company.

The estimated price of Alphabet is \$208,96 and the estimated price of Netflix is \$310,36, which goes hand to hand with the assessment done using the valuation by comparables method. The prices in this case are slightly higher but they still indicate that Alphabet's stock might be undervalued and Netflix's on the other hand might be overvalued.

Bond Valuation

In this section, we evaluate the bond prices of Alphabet and Netflix, while determining whether the prices are in line with market expectations.

For Alphabet, we choose a semi-annual bond with 5-year maturity and using the company's credit rating (Aa2 – Moody's) we get the spread which will be added to the current yield of Treasury bonds to deduce the yield to maturity (YTM) of the bond (5,31%). Next, the YTM is used to estimate the price of the bond (\$789,02) using the DCF model.

The same procedure is followed for Netflix, taking a semi-annual bond with a 10-year maturity, the spread is similarly acquired using the credit rating (Baa2) and the maturity, and added to the Treasury bonds' yield to calculate the YTM used later to estimate the price of the bond (\$955,67). Alphabet's bond may be undervalued based on its high credit rating and low YTM compared to the estimated price. In contrast, Netflix's bond price appears to be more aligned with market expectations considering its coupon rate and credit risk. However, these assessments must be taken in the context of the current interest rate environment and investor sentiment towards corporate bonds, especially in the technology and media sectors. Any recent changes in the companies' creditworthiness, market conditions, or macroeconomic factors could significantly influence these bond valuations.

Beta Analysis

In our analysis of Alphabet and Netflix, we computed the beta values of both firms using daily and monthly stock prices over the last five years.

	Daily Beta	Monthly Beta
Alphabet inc. (GOOGL)	1,152	1,052
Netflix (NFLX)	1,049	1,214

[Figure 1](#): Beta values of each company

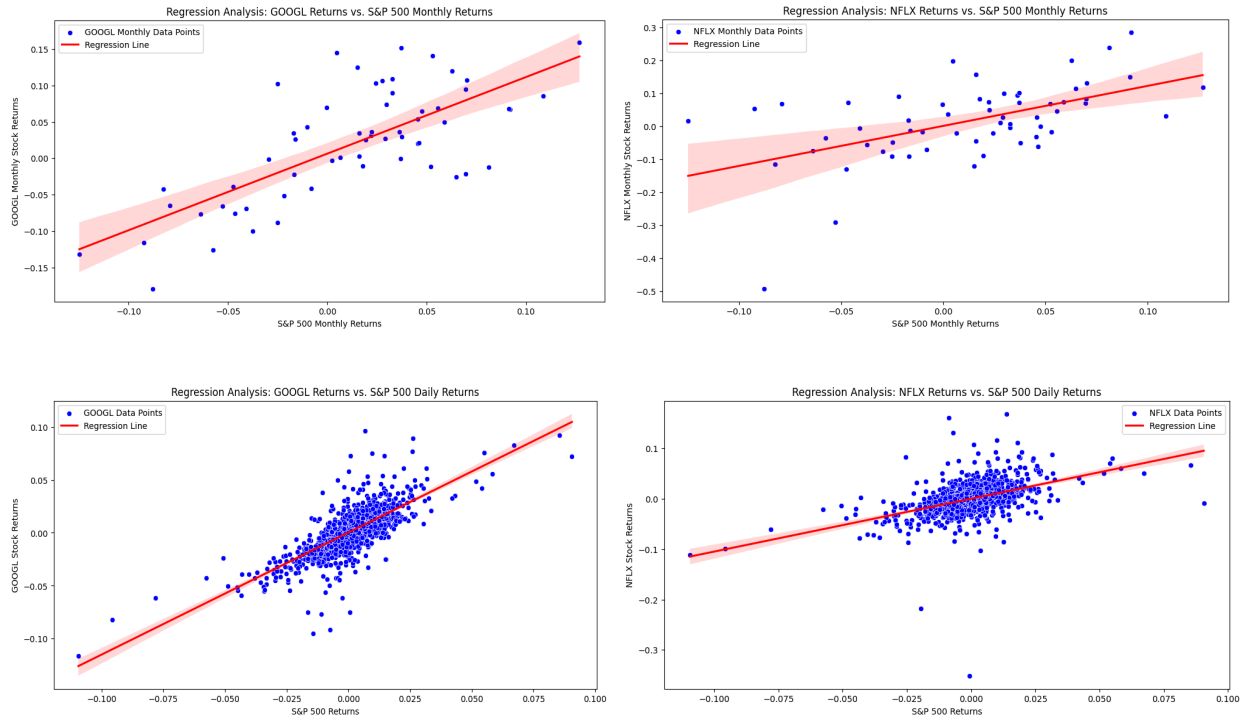


Figure 2: Scatterplots of reuters of companies vs index (monthly and daily)

Typically, we anticipate Netflix, with its focus on high growth in the streaming sector, to have a higher beta, reflecting greater volatility. However, our analysis reveals that Alphabet had a higher daily beta than Netflix.

Alphabet may have experienced specific occurrences that unusually affected its daily stock volatility over the last five years. The pandemic's differing impact on the volatility of daily stock prices could explain the variation in beta values. While Alphabet grappled with multifaceted market shifts, Netflix potentially benefited from increased demand for streaming services during lockdowns, which may have introduced a stabilizing effect on its stock volatility.

On the other hand, the monthly beta tends to smooth out short-term volatility and may better reflect the longer-term market risks associated with the firm.

Portfolio Construction and Analysis

In the following section, we form a portfolio with the two firms, and we assess how it would be in terms of risk-return, compared to each single stock.

We proceed by building 20000 portfolios with the following assumptions:

- All positions are long, representing a direct investment in the stocks.
- Weights must sum up to one, ensuring full capital allocation.

The evaluation is based on the daily stock price movements of Netflix and Alphabet over the last five years. For each portfolio, we calculate the expected return and the risk (standard deviation) to get the following efficient frontier:

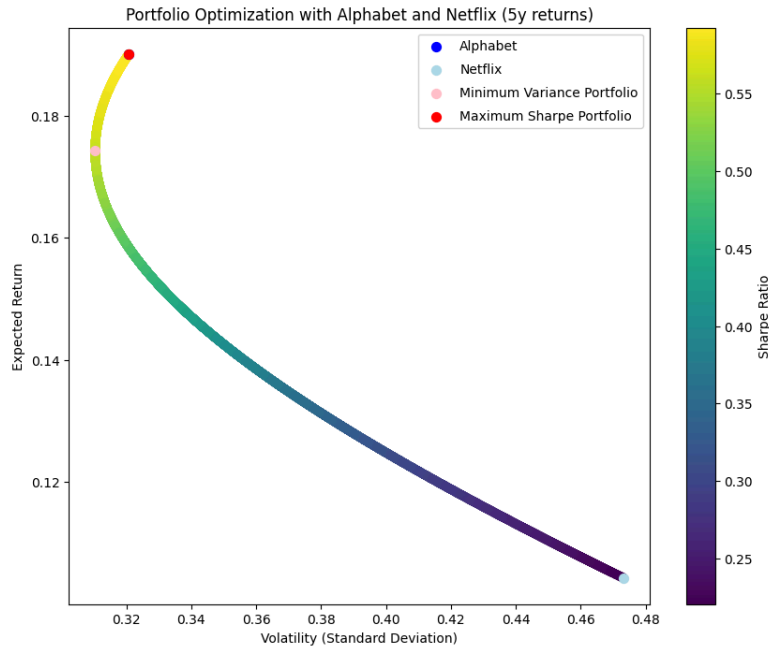


Figure 3: Efficient frontier using returns of last five years

The table below shows the risks and returns for each investment option:

Taking 5y Returns	Return	Risk (std.dev)
Alphabet inc. (GOOGL)	19,01%	32,09%
Netflix (NFLX)	10,43%	47,32%
Minimum Variance Portfolio	17,42%	31,04%
Maximum Sharpe Portfolio	19,01%	32,08%

Figure 4: Table showcasing risks and returns of investments (5y)

The minimum variance portfolio, emphasizing risk aversion comprises 81% Alphabet stock and 19% Netflix stock. On the other hand, the portfolio maximizing Sharpe Ratio, providing an optimal risk-return trade-off, indicates a full investment on Alphabet's stock, which means by investing only in Alphabet we can maximize the risk-return ratio, with only a marginal increase in risk for higher returns compared to the minimum variance portfolio. This is understandable since Alphabet's stock grants higher returns with lower risks considering the last 5 years returns, which also explains the unusual shape of the efficient frontier. The profound impact of the COVID-19 pandemic (Ashraf, 2020), particularly on Netflix, justifies the anomalous shape observed in the five-year efficient frontier. Netflix's stock experienced significant fluctuations and adverse performance during the pandemic, reflected in heightened volatility and negative returns. To further investigate these results, we zoom into a shorter time frame of one year (from 11-04-2023 to 11-04-2024) and we repeat the same process. We got the following results:

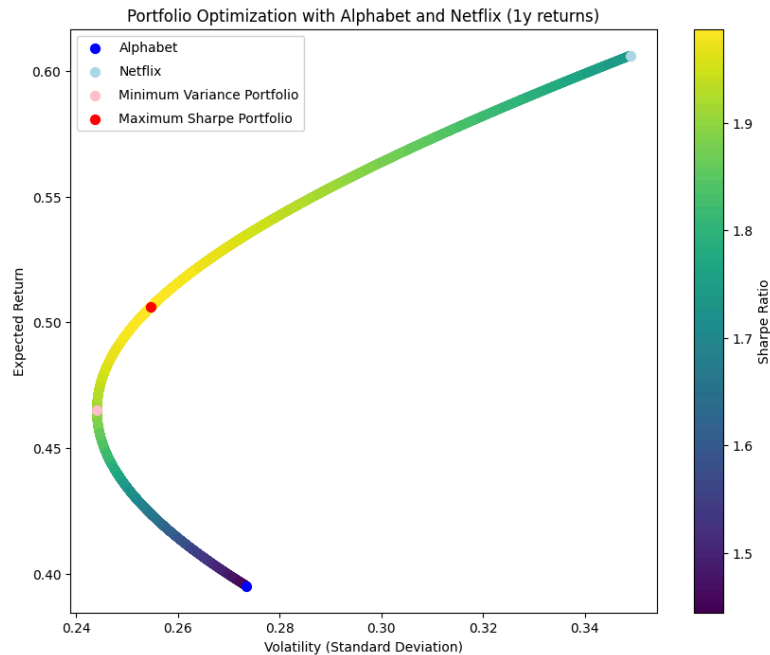


Figure 5: Efficient frontier using returns of the last year only

Taking 1y Returns		
	Return	Risk (std.dev)
Alphabet inc. (GOOGL)	39,50%	27,35%
Netflix (NFLX)	60,62%	34,90%
Minimum Variance Portfolio	46,49%	24,41%
Maximum Sharpe Portfolio	50,60%	25,46%

Figure 6: Table showcasing risks and returns of investments (1y)

We get a regular shape for the efficient frontier, and Netflix's expected returns are higher with higher volatility compared to Alphabet. The minimum variance allocates 67% for Alphabet and 33% for Netflix, achieving the lowest possible risk. The portfolio maximizing Sharpe Ratio, with a distribution of 47% Alphabet and 53% Netflix, aims for a higher expected return, approximately 4% greater, while incurring only a marginal 1% uptick in risk compared to the minimum variance portfolio. For investors willing to accommodate greater exposure to risk, it is ideal to invest in a portfolio maximizing Sharpe Ratio. Compared to this, a solitary investment Netflix's stock can bring 10% higher expected returns but with 10% higher risk, and Alphabet's stock on the other hand yields lower returns for higher risks, laying on the inefficient part of the frontier, therefore investing only on Alphabet wouldn't be a strategic decision given the data. Starting with the minimum variance portfolio, an investor can choose, depending on his risk adversity, by how much to increase the weight of Netflix's stock, increasing both risk and expected returns.

Investors may choose to calibrate their portfolios starting from the baseline of the minimum variance composition. Depending on individual risk tolerance and return objectives, one could

judiciously increase the proportion of Netflix's stock, thereby elevating both the potential risk and expected returns. This adjustment allows investors to navigate their risk-return preferences with precision and tailor their investment approach to align with their unique financial profiles and goals.

Capital Structure and Cost of Capital

Alphabet has \$110,916 billion in total assets, indicating the scale of the company, with a quite high equity base at \$283,379 billion suggesting a strong balance sheet, supported largely by retained earnings and common stock. The company has \$28,867 billion in total debt, with a small portion as short-term debt. The Debt-to-Equity Ratio (D/E) of 0.102, suggests that Alphabet has a very low level of debt relative to its equity, indicating a strong equity position and a low reliance on debt financing, with operations being heavily financed by shareholders' equity as suggested by the high equity ratio of 2,555. Moreover, the Debt-to-Asset Ratio of 0.26 shows that a quarter of the company's assets are financed through debt.

On the other hand, Netflix has \$48,732 billion in total assets, smaller than Alphabet, as expected given the nature of their respective businesses.

With a total equity of \$20,588 billion, Netflix has a smaller equity base than Alphabet and its total debt stands at \$16,973 billion. A higher D/E ratio of 0.824 suggests a more aggressive use of debt in its capital structure compared to Alphabet, and the equity ratio of 0.422 indicates a lower proportion of equity financing relative to Alphabet. The Debt-to-Asset Ratio of 0.3483 shows that nearly a third of Netflix's assets are financed by debt, which is higher than Alphabet's ratio.

With the cost of debt at 1.07%, Alphabet is benefiting from very low borrowing costs, likely due to its strong creditworthiness. While Netflix's cost of debt, at 4.12%, is significantly higher than Alphabet's, reflecting a higher risk premium demanded by lenders.

The cost of equity in Alphabet is 9.93%, influenced by its beta of 1.343, which suggests that Alphabet's stock is somewhat more volatile than the overall market.

On the other hand, Netflix has a slightly higher cost of equity at 10.53% and a beta of 1.494, indicating higher volatility and risk compared to Alphabet.

The WACC of 9.65% (Alphabet) shows the average rate that Alphabet is expected to pay on its combined debt and equity financing, accounting for the tax shield from debt. Netflix's WACC is marginally higher at 10.14%, which can be attributed to its higher cost of debt and equity.

Alphabet uses debt in a conservative way, maintaining a capital structure heavily relying on equity. This suggests a lower risk profile and a potential for lower costs of future financing. Netflix has a more aggressive approach to debt, with higher leverage ratios indicating a willingness to use debt for growth. This could suggest a higher potential return on equity, but it also comes with higher financial risk, as reflected in its higher WACC compared to Alphabet. Ultimately, investors might view Alphabet's financial stability and lower cost of capital as indicative of a safer investment profile, particularly attractive to risk-averse individuals. On the

other hand, Netflix's approach could be appealing to those seeking higher growth potential and willing to accept the accompanying higher risk.

Dividend Policy

Both companies don't pay any dividends and are not expected to pay in the future. With a plowback ratio of 1, the firms like to re-invest all their earnings for the growth of their respective businesses. These firms are focused more on long-term shareholder value generation.

Conclusion

Both Alphabet and Netflix present good investment profiles but have distinct business models, financial strategies, and market dynamics. Alphabet offers a more conservative investment option with its strong financial health and diversified business operations, providing stable, maybe more modest returns over time compared to Netflix. On the other hand, Netflix is more appealing for investors looking for higher growth potential, which is reflected in its aggressive strategy and international expansion. Netflix is financially more risky due to its bigger reliance on debt financing. Investors should weigh these factors based on their individual risk tolerance to make informed decisions aligned with their financial goals.

References

- Ashraf, B., 2020. Economic impact of government interventions during the COVID-19 pandemic: International evidence from financial markets. *Journal of Behavioral and Experimental Finance*, 27, pp. 100371 - 100371.
<https://doi.org/10.1016/j.jbef.2020.100371>.
- Bloomberg, n.d. Financial statements for Alphabet Inc, Netflix Inc. Available at: Bloomberg database (Accessed 11 April 2024). Financial data for computation purposes.
- CompaniesMarketCap (n.d.). Companies ranked by Market Cap - Available at: <https://companiesmarketcap.com/>
- Goldstein, I., Koijen, R., & Mueller, H., 2021. COVID-19 and Its Impact on Financial Markets and the Real Economy. *The Review of Financial Studies*.
<https://doi.org/10.1093/rfs/hhab085>.
- Yahoo Finance. (2024). Alphabet's (NASDAQ:GOOGL) investors will be pleased with their stellar 171% return over the last five years. [online] Available at: <https://finance.yahoo.com/news/alphabets-nasdaq-googl-investors-pleased-130027009.html>
- Yahoo Finance. (2024). Netflix Co-CEO Gregory Peters Sells 5,821 Shares. [online] Available at: <https://finance.yahoo.com/news/netflix-co-ceo-gregory-peters-081350449.html>
- Yahoo Finance, (2024). Historical stock prices for Alphabet Inc. (GOOGL), Netflix Inc. (NFLX), and the S&P 500 index. Available at: <https://finance.yahoo.com/> (Accessed 11 April 2024).

Appendix

The following GitHub repository <https://github.com/chakir7769/finance> contains an excel file, a jupyter notebook and the figures shown in this report in a separate folder.

The excel file, named 'FinanceProject.xlsx', contains all the computations mentioned in this report and the tables displayed with a detailed description of how they were performed.

The jupyter notebook, named 'fin.ipynb', contains the generated portfolios, the efficient frontier plots, the optimal portfolios and the beta calculations using the regression analysis with the description of how these calculations are performed.